Exercise 4.5

$$\left[\begin{aligned} a^3+b^3 &= (a+b)(a^2-ab+b^2) \\ a^3-b^3 &= (a-b)(a^2+ab+b^2) \end{aligned}\right]$$

1) (i) $8x^3 + y^3$

$\Rightarrow (2x)^3 + y^3$

$\Rightarrow (2x+y)((2x)^2 - 2xy + y^2)$

$\Rightarrow (2x+y)(4x^2 - 2xy + y^2)$

(ii) $64x^3 - 125y^3$

$\Rightarrow (4x)^3 - (5y)^3$

$\Rightarrow (4x - 5y)(16x^2 + 25y^2 - 20xy)$

2) (ii) $7a^3 + 56b^3$

$\Rightarrow 7(a^3 + 8b^3)$

$\Rightarrow 7((a)^3 + (2b)^3)$

$\Rightarrow 7(a+2b)(a^2 + (2b)^2 - 2ab)$

$\Rightarrow 7(a+2b)(a^2 + 4b^2 - 2ab)$

3) (ii) $8x^3 - \frac{1}{27y^3}$

$\Rightarrow (2x)^3 - \frac{1}{(3y)^3}$

$\Rightarrow (2x)^3 - \left(\frac{1}{3y}\right)^3$

$\Rightarrow \left(2x - \frac{1}{3y}\right)\left((2x)^2 + \left(\frac{1}{3y}\right)^2 + 2x \times \frac{1}{3y}\right)$

$\Rightarrow \left(2x - \frac{1}{3y}\right)\left(4x^2 + \frac{1}{9y^2} + \frac{2x}{3y}\right)$

4) (ii) $32x^4 - 500x$

$\Rightarrow 4x(8x^3 - 125)$

$\Rightarrow 4x((2x)^3 - 5^3)$

$\Rightarrow 4x(2x - 5)(4x^2 + 25 + 10x)$

5) (ii) $27(x+y)^3 + 8(2x-y)^3$

$\Rightarrow 3^3(x+y)^3 + 2^3(2x-y)^3$

$\Rightarrow (3(x+y))^3 + (2(2x-y))^3$

$\Rightarrow [(3(x+y)) + (2(2x-y))][3^2(x+y)^2 - 3(x+y)(2x-y) + 2^2(2x-y)^2]$

$\cancel{\Rightarrow [3(x+y) +}$

$\Rightarrow (3x+3y+4x-2y)[9(x+y)^2 - 3(x+y)(2x-y) + 4(2x-y)^2]$

$\Rightarrow (7x+y)[9(x^2+y^2+2xy) - 6(2x^2 - xy + 2xy - y^2) + 4(4x^2+y^2-4xy)]$

$\Rightarrow (7x+y)[9x^2+9y^2+18xy - 12x^2 - 6xy + 6y^2 + 16x^2 + 4y^2 - 16xy]$

$\Rightarrow (7x+y)[13x^2 - 4xy + 19y^2]$

6) (ii) $a^3 - b^3 - a + b$

$\Rightarrow (a^3 - b^3) - (a - b)$

$\Rightarrow (a - b)(a^2 + b^2 + ab) - (a - b)$

$\Rightarrow (a - b)(a^2 + ab + b^2 - 1)$

7) (ii) $a^3 - a - 120$

$\Rightarrow a^3 - 125 - a + 5$

$\Rightarrow (a^3 - 125) - (a - 5)$

$\Rightarrow (a^3 - 5^3) - (a - 5)$

$\Rightarrow (a - 5)(a^2 + 5^2 + 5a) - (a - 5)$

$\Rightarrow (a - 5)(a^2 + 5a + 25) - (a - 5)$

$\Rightarrow (a - 5)[a^2 + 5a + 25 - 1]$

8) (ii) $a^3 - 3a^2b + 3ab^2 - 2b^3$

$\Rightarrow (a - b)^3 - b^3$

$\Rightarrow (a - b)((a - b)^2 + (a - b)b + b^2)$

$\Rightarrow (a - b)(a^2 + b^2 - 2ab + ab - b^2 + b^2)$

$\Rightarrow (a - b)(a^2 + b^2 - ab)$

9) (ii) $a^3 - \frac{1}{a^3} - 2a + \frac{2}{a}$

$\Rightarrow a^3 - 2a + a - a + \frac{2}{a} + \frac{1}{a} - \frac{1}{a} - \frac{1}{a^3}$

$\Rightarrow a^3 - 3a + a\frac{3}{a} - \frac{1}{a} - \frac{1}{a^3}$

$\Rightarrow a^3 - \left(3 \cdot a^2 \times \frac{1}{a}\right) + \left(3 \times a \times \frac{1}{a^2}\right) - \frac{1}{a^3} + \left(a - \frac{1}{a}\right)$

$\Rightarrow \left(a - \frac{1}{a}\right)\left[\left(a - \frac{1}{a}\right)^2 + 1\right]$

$\Rightarrow \left(a - \frac{1}{a}\right)\left[a^2 + \frac{1}{a^2} - \left(2 \times a \times \frac{1}{a}\right) + 1\right]$

$\Rightarrow \left(a - \frac{1}{a}\right)\left(a^2 + \frac{1}{a^2} - 1\right)$

(10) (ii) $x^6 - 1$

$\Rightarrow (x^2)^3 - 1^3$

$\Rightarrow (x^2 - 1)((x^2)^2 + x^2 + 1^2)$

$\Rightarrow (x^2 - 1)(x^4 + x^2 + 1)$

11) (ii) $x^2 - \frac{8}{x}$

$\Rightarrow \left(x^2 - \frac{8}{x}\right)\frac{x}{x}$

$\Rightarrow x\left(x^2 - \frac{8}{x}\right) \times \frac{1}{x}$

$\Rightarrow (x^3 - 2^3) \times \frac{1}{x}$

$\Rightarrow \frac{1}{x}(x-2)(x^2 + 2x + 4)$

12) (ii) $32a^2x^3 - 8b^2x^3 - 4a^2y^3 + b^2y^3$

$\Rightarrow 8x^3(4a^2 - b^2) - y^3(4a^2 - b^2)$

$\Rightarrow (8x^3 - y^3)(4a^2 - b^2)$

$\Rightarrow ((2x)^3 - y^3)((2a)^2 - b^2)$

$\Rightarrow [(2x - y)((2x)^2 + 2xy + y^2)](2a + b)(2a - b)$

$\Rightarrow (2a + b)(2a - b)(2x - y)(4x^2 + 2xy + y^2)$

13) (ii) $x^6 - 7x^3 - 8$

$\Rightarrow x^6 - 7x^3 - 8 - x^3 + x^3$

$\Rightarrow x^6 - 8x^3 + x^3 - 2^3$

$\Rightarrow ((x^2)^3 - (2x)^3) + (x^3 - 2^3)$

$\Rightarrow (x^2 - 2x)(x^4 + 2x^3 + 4x^2) + (x - 2)(x^2 + 2x + 4)$

$\Rightarrow (x^2 - 2x)x^2(x^2 + 2x + 4) + (x - 2)(x^2 + 2x + 4)$